

Response Letter to Referees' Comments

1 Summary

We thank the Editor and the referee for their thoughtful, constructive comments. The manuscript has been revised accordingly with the following major changes:

1. The literature review section has been revised to incorporate the work suggested in the comments. A detailed discussion has been added to compare the proposed method with those referred to by the editor. More emphasis has been put on the distinctive elements of this paper, which are the shape restrictions and the adaptability of variance in response to either the conditional mean or marginal mean. For the inference method, we highlighted its computational ease without requiring a closed-form solution.
2. In the inference section, we referenced the resampling-subject bootstrap (RSB) method mentioned in the comments, and added a detailed discussion around the similarity and difference with the parametric bootstrap method employed in this paper. Our parametric bootstrap method shares the same logic as the RSB when maintaining the clustered structure in the data, but with a distinction - the bootstrap sample is regenerated using the estimated parameters instead of directly from the residuals. This choice stems from the parametric format in our model setting.
3. We have added clearer details about the selection of the degree of freedom K for the spline bases and the placement of internal knots. The degree of freedom K will be selected by AIC/BIC, and knots are evenly spaced.

Point-by-point responses are as follows, with the reviewer/editor's comments in *italic*.

2 To Referee 1

For a monotone function of a combination convexity and concavity for different interval, it seems that your method can only use I-splines. Is there a way to find the change points so that you could fit for different intervals with C-splines while preserve smoothness and monotonicity at the same time?

This observation holds significant value and could be considered for future exploration. In our simulation study, the third setting (function g_3) has intervals with both convex and concave patterns. While we solely employed monotone restrictions for this scenario, there's potential for improved estimation by incorporating the detection of change points. We have added a comment to the end of the last paragraph of Section 6.

Variances near boundaries are typically larger than that in the middle of a growth curve. The monotone restriction may increase/decrease the estimate of variance near a boundary. How do you control this?

Indeed this is a valid and valuable point for GCA. Our paper focused more on the situation that variance near one end of the boundary is larger. It can be the next step work to enable larger variance on both ends of the boundaries. We updated the writing in the second and fifth paragraph of Section 1 to emphasize that this is what motivated us to propose this method, and updated the last paragraph of Discussion section to include future work.

3 To Referee 2

1. The general setup as described in Section 2 appears to be very close to the ones in [Rice and Wu \(2001\)](#) and, for that matter, the models of [Huang et al. \(2002, 2004\)](#). Also, see the relevant chapters in the the book [Wu and Tian \(2018\)](#). The only difference is the M-splines, I-splines, C-splines, etc., as opposed to the usual B-splines. To clearly state the objective of the paper, it is better to state the overall model before the spline basis expressions. You need to inform the reader which functions should be approximated by splines, why M-splines, I-splines, C-splines, etc. are needed, and pros and cons of these base choices. For clarity, please also state and justify the similarities and differences from the usual polynomial splines (see [Wu and Tian \(2018\)](#)).

The literature highlighted by the editor holds great relevance to our work, prompting us to enhance the literature review. We updated the third paragraph of Section 1, the first and second paragraphs of Section 2 to include the following change: literatures were added to include past work around using splines on clustered or longitudinal data [Rice and Wu \(2001\)](#); [Huang et al. \(2002, 2004\)](#); [Wu and Tian \(2018\)](#). We also added more emphasis to make it clear that the noval part of this paper is the shape restrictions on both mean and variation terms. Please see the fifth paragraph of Section 1.

2. Section 3.1: Model (3.1) appears to be a usual LME with some basis variance functions. This setup is more restrictive compared to the ones in [Rice and Wu \(2001\)](#) or [Huang et al. \(2002\)](#) or the models in [Wu and Tian \(2018\)](#). In addition, the authors did not specify K to be dependent on the sample size n . For growth curves (the topic of the paper), a classical LME is clearly not appropriate. It will be more interesting and relevant if β is allowed to be a function of t . It will dramatically reduce the value of this paper if your general setup is a standard LME (3.1). In addition, growth curves in practice, such as children's growth, e.g. height, weight, body mass index, blood pressure and cholesterol, over age, may not be a linear model, and is more appropriate to be nonparametric. I actually think that you can remove this subsection completely and do straight to the longitudinal data case in Section 3.2.

It is a great point that clustered data is the more prevalent scenario in GCA, and starting with the independent data and model in this paper might be confusing. We updated the first and second paragraphs of Section 3 to add more clarity on the reason to include independent data scenario: the main focus is to use shape restrictions on GCA, and we want the paper to be thorough and covers both scenarios since certain shape restrictions may occur to either of them. This perspective is also reflected in the pancrea data analysis presented in Section 5.1.

It is important to acknowledge that the mean pattern of GCA is not linear with time. We added a second paragraph of Section 3.1 to show it more clearly that the spline bases of

time should be used in the mean model. The inference process won't change when getting the coefficient estimation of coefficients of spline bases, and the model selection method to choose the degree and degree of freedom of spline bases are the same as for the error variance.

Regarding the spline bases, the degree of freedom K will be chosen using the model selection method outlined in Section 3.4. We updated section 3.1 and 3.2 to emphasize this when we first introduced K . Additionally, we've made updates to the first and second paragraphs of Section 3.4, underscoring that the choice of K should adapt with the sample size, both overall (n) and within clusters (n_i).

Recognizing the valuable consideration of permitting the coefficient β to evolve with time in GCA, our model is designed to effectively manage this scenario. This can be accomplished through the incorporation of interaction terms between covariates and the spline bases of time within the mean function. We updated the last paragraphs of Section 3.1 and 3.2 to reflect this, with the two references cited (Huang et al., 2002, 2004).

3. Section 3.2: It is better to start with the framework here. Section 3.1 adds little value, and may be removed to make the paper more succinct.

Please refer to the response to comment 2, first paragraph.

4. Sections 3.1 & 3.2: You need to define your data clearly. Such as, what are $\mathbf{Y}_i$, $\mathbf{t}_i$, etc.? It is better to define the structure of your data at the beginning of this section. The number of repeated measurements n_i may or may not be the same for each subject i , and n_i may or may not be large. You need to make all these assumptions clear and reasonable for your real data examples.

In consideration of valuable feedback, we've made adjustments to the writing to improve the clarity surrounding data definitions and notations. Kindly refer to the refined explanations in the first paragraphs of both Section 3.1 and 3.2. Furthermore, we have updated the notations in model (3.2) to provide a clearer depiction of the underlying settings.

5. Algorithm 1, page 4: Huang et al. (2002, 2004) used OLS or weight adjusted by n and n_i (see, Wu and Tian (2018), Section 9.2). Is your iterative Algorithm 1 better than the OLS or weighted LS of Huang, Lan and Wu?

This is a very helpful point to make our discussion more thorough. We've revisited the writing to provide clearer explanations for using the iterative algorithm, as outlined in the first paragraph of Section 3.3. The primary rationale behind this choice lies in the simplicity and efficiency of computation, avoiding the need for intricate closed-form solutions. This method exhibits flexibility and ease of application across various scenarios, particularly in cases where deriving closed-form solutions proves challenging—such as situations when error variance changes with conditional or marginal mean.

6. Equation (3.2) of page 4: Same as above, this model is a slight generalization of the classical LME, and is very restrictive and may not be appropriate for growth curves. Only adding a basis function with fixed K for variance is not sufficient. You need to consider a more practical model that people can justify. In my view, a generalization of the setup of Huang et al. (2002) with basis approximated variances is a more relevant and interesting model.

Please see the detailed response to comment 1 and 2. We've updated the manuscript according to the valuable comments to make the discussion more thorough and solid.

7. Section 3.3: The likelihood based inference, although popular, is not suitable for spline based models, because you really should allow K to change with your sample, i.e. n and n_i . Existing inferences, such as [Huang et al. \(2002\)](#) and [Wu and Tian \(2018\)](#) used "resampling-subject bootstrap" (RSB) inferences. I wonder how your Algorithm 2 is compared with the "resampling-subject bootstrap." It appears that the RSB can be more generally applied. In any case, you need to justify the use of your algorithm over the RSB.

It is a great point that we should compare the proposed parametric bootstrap method with the RSB in existing literature, since they share a lot of similarities. We updated the second paragraph of Section 3.3 to include the comparison in writing. The key distinction lies in generating the bootstrapped error term using the estimated parameters instead of residuals. This is due to our ability to estimate the error variance using spline bases.

8. Section 3.4: The use of AIC and BIC appears to be a generalization of the ones in [Rice and Wu \(2001\)](#). You may need to clarify what is novel in your setting here.

It is a valuable point that we should cite past works that uses same method as a support. We updated the first and second paragraphs of Section 3.4 to show that the AIC and BIC criteria align with the framework presented in [Rice and Wu \(2001\)](#). The only difference is when numerical integration becomes necessary for certain scenarios when closed-form integration is unavailable.

9. It is not clear how the numbers and locations of the knots are selected. In [Rice and Wu \(2001\)](#), [Huang et al. \(2002\)](#), and [Wu and Tian \(2018\)](#), the splines are selected by 'deleting subject cross validation.' Please provide a procedure for smoothing parameter selection here.

We appreciate the comment and incorporate it to add more clarity on the writing. In the update to Section 3.4, we've placed additional emphasis on the selection of knots, drawing a comparison with the 'deleting subject cross-validation' method as detailed in the paper by [Rice and Wu \(2001\)](#). Furthermore, we've revisited the paragraphs discussing Model (3.1) and Model (3.2) to underscore that the choice of K should be made through the model selection method outlined in Section 3.4.

10. Simulation: I am fine with your simulation. But, you may want to remove Section 4.1 for the i.i.d. data case, and change the setting to a more general model in Section 2. The LME data case is really not adding anything new to the literature. In particular, you should not assume the number of spline terms K to be fixed. In practice, it should be selected.

Please see the response to comment 1 and comment 2.

11. Section 5: Please give a clear but brief description of the datasets, including citing the studies, whether the data is available, how to obtain the data, etc. We want to make sure that the information (data and publications) is available to our readers and the biomedical conclusion is meaningful.

Additional information has been incorporated into the second paragraph of Section 5.1 and the first paragraph of Section 5.2.

12. Section 5.1, equation for fetal pancreas length: How do you select the number and location of the knots for t ? What is your justification? It appears that you used BIC to select K_1 and K_2 . What are the medical implications and conclusions of your study? What new information does your study provide as opposed to the published results?

Please see the response to comment 2.

13. The comments above, #12, also apply to the "Chicken Weight" example of Section 5.2.

In the updated manuscript, we've extended the discussion in the final paragraph of Section 5.2 to include further insights regarding the incorporation of new values in real-life scenarios.

14. It will be helpful to add an appendix with R code. It will be tremendously helpful to our readers.

The R code is available on a public github repo, and the link is provided in the Section 1 Introduction.

References

- Huang, J. Z., C. O. Wu, and L. Zhou (2002). Varying-coefficient models and basis function approximations for the analysis of repeated measurements. *Biometrika* 89(1), 111–128.
- Huang, J. Z., C. O. Wu, and L. Zhou (2004). Polynomial spline estimation and inference for varying coefficient models with longitudinal data. *Statistica Sinica* 14, 763–788.
- Rice, J. A. and C. O. Wu (2001). Nonparametric mixed effects models for unequally sampled noisy curves. *Biometrics* 57(1), 253–259.
- Wu, C. O. and X. Tian (2018). *Nonparametric Models for Longitudinal Data: with Implementation in R*. CRC Press.